Renewable Energy AI - Operational Report

Run ID: 8092c3bc-38e4-4c14-bd44-856ce295f39f
Generated UTC: 2026-10-12T12:38:15.807192+00:00
Report period (UTC): 2026-10-12 through 2026-10-14
Run recorded: 2026-10-12T12:20:51.183048Z through 2026-10-12T12:20:51.183048Z

Executive summary

Workflow state: awaiting human review. Plan version: 1.
Human approval: pending; persisted decisions in period: 0.
Stored measurements in period: 0; included in report: 0 (maximum 100 rows).
Generation/demand forecast intervals in period: 48/48.
Optimization intervals in period: 48; aligned measured comparisons: 0.
Optimization output is a recommendation. No physical grid dispatch is represented by this report.

Workflow status and human review

Effective workflow state: awaiting_human_review
Approval state: pending
Automated review: ready_for_human_review
Approval history: no human decision is recorded in the selected period.

Registered assets and stored measurements

Asset registry reflects current database records; historical registry snapshots are not stored.
Solar Farm Alpha | solar | 10.00 MW | active | Hyderabad
asset | wind | 0.31 MW | inactive | hyderabad
wind test | wind | 2.01 MW | active | Hyderabad
asset 4 | battery | 2.71 MW | active | Hyderabad
No stored measurements fall within the selected period.

Weather inputs and provenance

Provider: Open-Meteo forecast API.
Retrieved: 2026-10-12T12:20:41.369574Z.
Fallback status: not_used.
Current weather input: time=2026-10-12T12:15, interval=900, temperature_2m=31.2, relative_humidity_2m=31, precipitation=0.0, cloud_cover=0, wind_speed_10m=3.76, wind_direction_10m=30, wind_speed_100m=5.59, wind_direction_100m=32, shortwave_radiation=33.0, direct_radiation=12.0.
Timestamped weather forecast records included in period: 48 of 48.

Solar and wind forecasts

Model source: Synthetic training dataset.
Model version: {"solar":"7ad4557981ac","wind":"fe254e57cd96"}.
Provenance limitation: model training data is synthetic; these metrics do not establish real-world forecast accuracy.

Stored held-out evaluation output (model evaluation, not period-filtered):
Intervals included: 48 of 48 stored forecast rows.

Demand forecast

Model source: synthetic_energy.csv.
Model version: 086525813a1c.
Provenance limitation: model training data is synthetic; these metrics do not establish real-world forecast accuracy.
Stored held-out evaluation output (model evaluation, not period-filtered):
model_mae_mw: 0.2254 MW
model_rmse_mw: 0.2768 MW
same_hour_previous_day_mae_mw: 0.2983 MW
same_hour_previous_day_rmse_mw: 0.3671 MW
holdout_records: 428.0000
holdout_start_utc: 2026-09-23T17:00:00+00:00
holdout_end_utc: 2026-10-11T12:00:00+00:00
evaluation_data_source: synthetic_energy.csv
evaluation_scope: Chronological held-out synthetic-facility observations
Intervals included: 48 of 48 stored forecast rows.

Energy balance and optimization recommendation

Optimizer status: Optimal; feasible: True.
Objective: Prioritize unmet-demand reduction, then grid-import reduction, renewable-curtailment reduction, and lower cycling..
Optimizer objective score for the full workflow horizon (unitless): 135323.83.
Selected-period totals recomputed from timestamped schedule rows in this UTC reporting period:
Grid imports 135.32 MWh | exports 21.73 MWh | unmet demand 0.00 MWh | curtailed generation 0.00 MWh | efficiency losses 1.314 MWh.
Battery: capacity 8.00 MWh; initial SoC 45.00%; terminal SoC 45.00%.
Full-horizon constraint battery_capacity_mwh: 8.0
Full-horizon constraint max_charge_mw: 2.0
Full-horizon constraint max_discharge_mw: 2.0
Full-horizon constraint min_soc_pct: 10.0
Full-horizon constraint max_soc_pct: 90.0
Full-horizon constraint charge_efficiency: 0.95
Full-horizon constraint discharge_efficiency: 0.95
Full-horizon constraint initial_soc_pct: 45.0
Full-horizon constraint terminal_soc_pct: 45.0
Full-horizon constraint timestep_hours: 1.0
Schedule rows in selected period: 48 of 48.

Performance analysis and alerts

Measurement comparison status: no_aligned_measurements; aligned actual timestamps in period: 0.
Screening threshold: 0.78 MW.
No stored asset measurements align exactly with forecast timestamps.

No actual-versus-forecast deviations are reported because no aligned actual measurement is available for this period.
Findings: 26 evidence records across 1 finding category.
Generation Demand Imbalance (26): Forecast generation is below 25% of forecast demand.
Persisted alerts for this run in selected period: 27 records.
high Inactive Assets | 1 records | IDs 454 | created 2026-10-12T12:20:51.260805Z to 2026-10-12T12:20:51.260805Z | statuses open=1 | 1 asset(s) are marked inactive.
medium Generation Demand Imbalance | 26 records | IDs 455, 456, 457, 458, 459, 460, 461, 462, 463, 464, 465, 466, 467, 468, 469, 470, 471, 472, 473, 474, 475, 476, 477, 478, 479, 480 | created 2026-10-12T12:20:51.260805Z to 2026-10-12T12:20:51.260805Z | statuses open=26 | Forecast generation is below 25% of forecast demand.

Data coverage and limitations

1 finding had no usable timestamp and were excluded from the filtered finding detail.
Weather current conditions, run-level model evaluation metrics, and the current asset registry are snapshots without period-specific history; they are disclosed as such and are not presented as period-filtered observations.
Workflow status is the saved state of the selected run. Execution-trace events have no individual timestamps, so the full run trace is included and is not period-filtered.

Appendix: selected forecast and finding evidence

Solar/wind forecast coverage: 48 intervals from 2026-10-12T13:00:00Z to 2026-10-14T12:00:00Z.
Peak forecast solar: 8.91 MW at 2026-10-13T07:00:00Z; peak wind: 0.47 MW at 2026-10-14T12:00:00Z.
Demand forecast coverage: 48 intervals; peak 6.08 MW at 2026-10-13T19:00:00Z.
Demand lag-source counts: earlier_model_prediction=24, earlier_prediction_fallback=23, last_observed_value_fallback=1.

Appendix: timestamped optimization schedule

Power columns are MW; battery energy is MWh. One row is one stored interval.
2026-10-12T13:00:00Z | gen 0.48 | demand 5.24 | charge 0.00 | discharge 0.66 | import 4.11 | export 0.00 | unmet 0.00 | curtailed 0.00 | battery 2.91 MWh / 36.32% SoC
2026-10-12T14:00:00Z | gen 0.21 | demand 4.92 | charge 0.00 | discharge 2.00 | import 2.71 | export 0.00 | unmet 0.00 | curtailed 0.00 | battery 0.80 MWh / 10.00% SoC
2026-10-12T15:00:00Z | gen 0.20 | demand 5.10 | charge 0.00 | discharge 0.00 | import 4.90 | export 0.00 | unmet 0.00 | curtailed 0.00 | battery 0.80 MWh / 10.00% SoC
2026-10-12T16:00:00Z | gen 0.11 | demand 5.40 | charge 0.00 | discharge 0.00 | import 5.29 | export 0.00 | unmet 0.00 | curtailed 0.00 | battery 0.80 MWh / 10.00% SoC
2026-10-12T17:00:00Z | gen 0.13 | demand 5.79 | charge 0.00 | discharge 0.00 | import 5.65 | export 0.00 | unmet 0.00 | curtailed 0.00 | battery 0.80 MWh / 10.00% SoC
2026-10-12T18:00:00Z | gen 0.15 | demand 5.79 | charge 0.00 | discharge 0.00 | import 5.65 | export 0.00 | unmet 0.00 | curtailed 0.00 | battery 0.80 MWh / 10.00% SoC
2026-10-12T19:00:00Z | gen 0.11 | demand 5.98 | charge 0.00 | discharge 0.00 | import 5.87 | export 0.00 | unmet 0.00 | curtailed 0.00 | battery 0.80 MWh / 10.00% SoC
2026-10-12T20:00:00Z | gen 0.10 | demand 5.86 | charge 0.00 | discharge 0.00 | import 5.76 | export 0.00 | unmet 0.00 | curtailed 0.00 | battery 0.80 MWh / 10.00% SoC

2026-10-12T21:00:00Z | gen 0.10 | demand 5.53 | charge 0.00 | discharge 0.00 | import 5.43 | export 0.00 | unmet 0.00 | curtailed 0.00 | battery 0.80 MWh / 10.00% SoC
2026-10-12T22:00:00Z | gen 0.12 | demand 5.41 | charge 0.00 | discharge 0.00 | import 5.28 | export 0.00 | unmet 0.00 | curtailed 0.00 | battery 0.80 MWh / 10.00% SoC
2026-10-12T23:00:00Z | gen 0.14 | demand 4.94 | charge 0.00 | discharge 0.00 | import 4.80 | export 0.00 | unmet 0.00 | curtailed 0.00 | battery 0.80 MWh / 10.00% SoC
2026-10-13T00:00:00Z | gen 0.13 | demand 3.91 | charge 0.00 | discharge 0.00 | import 3.78 | export 0.00 | unmet 0.00 | curtailed 0.00 | battery 0.80 MWh / 10.00% SoC
2026-10-13T01:00:00Z | gen 0.25 | demand 4.01 | charge 0.00 | discharge 0.00 | import 3.76 | export 0.00 | unmet 0.00 | curtailed 0.00 | battery 0.80 MWh / 10.00% SoC
2026-10-13T02:00:00Z | gen 1.40 | demand 4.00 | charge 0.00 | discharge 0.00 | import 2.60 | export 0.00 | unmet 0.00 | curtailed 0.00 | battery 0.80 MWh / 10.00% SoC
2026-10-13T03:00:00Z | gen 3.65 | demand 4.21 | charge 0.00 | discharge 0.00 | import 0.56 | export 0.00 | unmet 0.00 | curtailed 0.00 | battery 0.80 MWh / 10.00% SoC
2026-10-13T04:00:00Z | gen 5.84 | demand 4.24 | charge 0.00 | discharge 0.00 | import 0.00 | export 1.60 | unmet 0.00 | curtailed 0.00 | battery 0.80 MWh / 10.00% SoC
2026-10-13T05:00:00Z | gen 7.60 | demand 4.39 | charge 0.00 | discharge 0.00 | import 0.00 | export 3.22 | unmet 0.00 | curtailed 0.00 | battery 0.80 MWh / 10.00% SoC
2026-10-13T06:00:00Z | gen 8.77 | demand 4.84 | charge 0.43 | discharge 0.00 | import 0.00 | export 3.50 | unmet 0.00 | curtailed 0.00 | battery 1.21 MWh / 15.17% SoC
2026-10-13T07:00:00Z | gen 9.23 | demand 5.20 | charge 2.00 | discharge 0.00 | import 0.00 | export 2.03 | unmet 0.00 | curtailed 0.00 | battery 3.11 MWh / 38.92% SoC
2026-10-13T08:00:00Z | gen 8.81 | demand 5.54 | charge 2.00 | discharge 0.00 | import 0.00 | export 1.27 | unmet 0.00 | curtailed 0.00 | battery 5.01 MWh / 62.67% SoC
2026-10-13T09:00:00Z | gen 7.72 | demand 5.70 | charge 2.00 | discharge 0.00 | import 0.00 | export 0.03 | unmet 0.00 | curtailed 0.00 | battery 6.91 MWh / 86.42% SoC
2026-10-13T10:00:00Z | gen 6.06 | demand 5.75 | charge 0.30 | discharge 0.00 | import 0.00 | export 0.00 | unmet 0.00 | curtailed 0.00 | battery 7.20 MWh / 90.00% SoC
2026-10-13T11:00:00Z | gen 4.06 | demand 5.52 | charge 0.00 | discharge 1.46 | import 0.00 | export 0.00 | unmet 0.00 | curtailed 0.00 | battery 5.66 MWh / 70.79% SoC
2026-10-13T12:00:00Z | gen 1.85 | demand 5.21 | charge 0.00 | discharge 2.00 | import 1.36 | export 0.00 | unmet 0.00 | curtailed 0.00 | battery 3.56 MWh / 44.47% SoC
2026-10-13T13:00:00Z | gen 0.32 | demand 5.07 | charge 0.00 | discharge 2.00 | import 2.75 | export 0.00 | unmet 0.00 | curtailed 0.00 | battery 1.45 MWh / 18.16% SoC
2026-10-13T14:00:00Z | gen 0.18 | demand 4.99 | charge 0.00 | discharge 0.62 | import 4.20 | export 0.00 | unmet 0.00 | curtailed 0.00 | battery 0.80 MWh / 10.00% SoC
2026-10-13T15:00:00Z | gen 0.19 | demand 5.11 | charge 0.00 | discharge 0.00 | import 4.92 | export 0.00 | unmet 0.00 | curtailed 0.00 | battery 0.80 MWh / 10.00% SoC
2026-10-13T16:00:00Z | gen 0.12 | demand 5.44 | charge 0.00 | discharge 0.00 | import 5.32 | export 0.00 | unmet 0.00 | curtailed 0.00 | battery 0.80 MWh / 10.00% SoC
2026-10-13T17:00:00Z | gen 0.10 | demand 5.48 | charge 0.00 | discharge 0.00 | import 5.38 | export 0.00 | unmet 0.00 | curtailed 0.00 | battery 0.80 MWh / 10.00% SoC
2026-10-13T18:00:00Z | gen 0.05 | demand 5.86 | charge 0.00 | discharge 0.00 | import 5.81 | export 0.00 | unmet 0.00 | curtailed 0.00 | battery 0.80 MWh / 10.00% SoC
2026-10-13T19:00:00Z | gen 0.04 | demand 6.08 | charge 0.00 | discharge 0.00 | import 6.04 | export 0.00 | unmet 0.00 | curtailed 0.00 | battery 0.80 MWh / 10.00% SoC
2026-10-13T20:00:00Z | gen 0.04 | demand 6.03 | charge 0.00 | discharge 0.00 | import 5.99 | export 0.00 | unmet 0.00 | curtailed 0.00 | battery 0.80 MWh / 10.00% SoC
2026-10-13T21:00:00Z | gen 0.10 | demand 5.58 | charge 0.00 | discharge 0.00 | import 5.49 | export 0.00 | unmet 0.00 | curtailed 0.00 | battery 0.80 MWh / 10.00% SoC

2026-10-13T22:00:00Z | gen 0.16 | demand 5.16 | charge 0.00 | discharge 0.00 | import 4.99 | export 0.00 | unmet 0.00 | curtailed 0.00 | battery 0.80 MWh / 10.00% SoC
2026-10-13T23:00:00Z | gen 0.22 | demand 4.97 | charge 0.00 | discharge 0.00 | import 4.75 | export 0.00 | unmet 0.00 | curtailed 0.00 | battery 0.80 MWh / 10.00% SoC
2026-10-14T00:00:00Z | gen 0.20 | demand 4.02 | charge 0.00 | discharge 0.00 | import 3.82 | export 0.00 | unmet 0.00 | curtailed 0.00 | battery 0.80 MWh / 10.00% SoC
2026-10-14T01:00:00Z | gen 0.23 | demand 3.99 | charge 0.00 | discharge 0.00 | import 3.76 | export 0.00 | unmet 0.00 | curtailed 0.00 | battery 0.80 MWh / 10.00% SoC
2026-10-14T02:00:00Z | gen 1.43 | demand 3.98 | charge 0.00 | discharge 0.00 | import 2.55 | export 0.00 | unmet 0.00 | curtailed 0.00 | battery 0.80 MWh / 10.00% SoC
2026-10-14T03:00:00Z | gen 3.61 | demand 4.22 | charge 0.00 | discharge 0.00 | import 0.61 | export 0.00 | unmet 0.00 | curtailed 0.00 | battery 0.80 MWh / 10.00% SoC
2026-10-14T04:00:00Z | gen 5.77 | demand 4.21 | charge 0.00 | discharge 0.00 | import 0.00 | export 1.56 | unmet 0.00 | curtailed 0.00 | battery 0.80 MWh / 10.00% SoC
2026-10-14T05:00:00Z | gen 7.45 | demand 4.40 | charge 0.00 | discharge 0.00 | import 0.00 | export 3.05 | unmet 0.00 | curtailed 0.00 | battery 0.80 MWh / 10.00% SoC
2026-10-14T06:00:00Z | gen 8.47 | demand 4.81 | charge 0.85 | discharge 0.00 | import 0.00 | export 2.81 | unmet 0.00 | curtailed 0.00 | battery 1.61 MWh / 20.13% SoC
2026-10-14T07:00:00Z | gen 8.97 | demand 5.21 | charge 2.00 | discharge 0.00 | import 0.00 | export 1.75 | unmet 0.00 | curtailed 0.00 | battery 3.51 MWh / 43.88% SoC
2026-10-14T08:00:00Z | gen 8.59 | demand 5.69 | charge 2.00 | discharge 0.00 | import 0.00 | export 0.90 | unmet 0.00 | curtailed 0.00 | battery 5.41 MWh / 67.63% SoC
2026-10-14T09:00:00Z | gen 7.64 | demand 5.90 | charge 1.74 | discharge 0.00 | import 0.00 | export 0.00 | unmet 0.00 | curtailed 0.00 | battery 7.07 MWh / 88.34% SoC
2026-10-14T10:00:00Z | gen 5.93 | demand 5.79 | charge 0.14 | discharge 0.00 | import 0.00 | export 0.00 | unmet 0.00 | curtailed 0.00 | battery 7.20 MWh / 90.00% SoC
2026-10-14T11:00:00Z | gen 3.77 | demand 5.40 | charge 0.00 | discharge 1.42 | import 0.21 | export 0.00 | unmet 0.00 | curtailed 0.00 | battery 5.71 MWh / 71.32% SoC
2026-10-14T12:00:00Z | gen 1.88 | demand 5.12 | charge 0.00 | discharge 2.00 | import 1.24 | export 0.00 | unmet 0.00 | curtailed 0.00 | battery 3.60 MWh / 45.00% SoC

Appendix: timestamped finding evidence

generation_demand_imbalance | 2026-10-12T13:00:00Z | Forecast shortfall 4.77 MW | severity medium | Forecast generation is below 25% of forecast demand.
generation_demand_imbalance | 2026-10-12T14:00:00Z | Forecast shortfall 4.71 MW | severity medium | Forecast generation is below 25% of forecast demand.
generation_demand_imbalance | 2026-10-12T15:00:00Z | Forecast shortfall 4.90 MW | severity medium | Forecast generation is below 25% of forecast demand.
generation_demand_imbalance | 2026-10-12T16:00:00Z | Forecast shortfall 5.29 MW | severity medium | Forecast generation is below 25% of forecast demand.
generation_demand_imbalance | 2026-10-12T17:00:00Z | Forecast shortfall 5.65 MW | severity medium | Forecast generation is below 25% of forecast demand.
generation_demand_imbalance | 2026-10-12T18:00:00Z | Forecast shortfall 5.65 MW | severity medium | Forecast generation is below 25% of forecast demand.
generation_demand_imbalance | 2026-10-12T19:00:00Z | Forecast shortfall 5.87 MW | severity medium | Forecast generation is below 25% of forecast demand.
generation_demand_imbalance | 2026-10-12T20:00:00Z | Forecast shortfall 5.76 MW | severity medium | Forecast generation is below 25% of forecast demand.
generation_demand_imbalance | 2026-10-12T21:00:00Z | Forecast shortfall 5.43 MW | severity medium | Forecast generation is below 25% of forecast demand.

generation_demand_imbalance | 2026-10-12T22:00:00Z | Forecast shortfall 5.28 MW | severity medium | Forecast generation is below 25% of forecast demand.
generation_demand_imbalance | 2026-10-12T23:00:00Z | Forecast shortfall 4.80 MW | severity medium | Forecast generation is below 25% of forecast demand.
generation_demand_imbalance | 2026-10-13T00:00:00Z | Forecast shortfall 3.78 MW | severity medium | Forecast generation is below 25% of forecast demand.
generation_demand_imbalance | 2026-10-13T01:00:00Z | Forecast shortfall 3.76 MW | severity medium | Forecast generation is below 25% of forecast demand.
generation_demand_imbalance | 2026-10-13T13:00:00Z | Forecast shortfall 4.75 MW | severity medium | Forecast generation is below 25% of forecast demand.
generation_demand_imbalance | 2026-10-13T14:00:00Z | Forecast shortfall 4.82 MW | severity medium | Forecast generation is below 25% of forecast demand.
generation_demand_imbalance | 2026-10-13T15:00:00Z | Forecast shortfall 4.92 MW | severity medium | Forecast generation is below 25% of forecast demand.
generation_demand_imbalance | 2026-10-13T16:00:00Z | Forecast shortfall 5.32 MW | severity medium | Forecast generation is below 25% of forecast demand.
generation_demand_imbalance | 2026-10-13T17:00:00Z | Forecast shortfall 5.38 MW | severity medium | Forecast generation is below 25% of forecast demand.
generation_demand_imbalance | 2026-10-13T18:00:00Z | Forecast shortfall 5.81 MW | severity medium | Forecast generation is below 25% of forecast demand.
generation_demand_imbalance | 2026-10-13T19:00:00Z | Forecast shortfall 6.04 MW | severity medium | Forecast generation is below 25% of forecast demand.
generation_demand_imbalance | 2026-10-13T20:00:00Z | Forecast shortfall 5.99 MW | severity medium | Forecast generation is below 25% of forecast demand.
generation_demand_imbalance | 2026-10-13T21:00:00Z | Forecast shortfall 5.49 MW | severity medium | Forecast generation is below 25% of forecast demand.
generation_demand_imbalance | 2026-10-13T22:00:00Z | Forecast shortfall 4.99 MW | severity medium | Forecast generation is below 25% of forecast demand.
generation_demand_imbalance | 2026-10-13T23:00:00Z | Forecast shortfall 4.75 MW | severity medium | Forecast generation is below 25% of forecast demand.
generation_demand_imbalance | 2026-10-14T00:00:00Z | Forecast shortfall 3.82 MW | severity medium | Forecast generation is below 25% of forecast demand.
generation_demand_imbalance | 2026-10-14T01:00:00Z | Forecast shortfall 3.76 MW | severity medium | Forecast generation is below 25% of forecast demand.

Alert record detail is summarized above by severity and reason; IDs, creation-time range, and status counts identify the persisted records.
Data rows without a timestamp and therefore excluded from this filtered appendix: forecasts/schedule 0; comparisons 0; findings 1; approval decisions 0.

Appendix: workflow execution trace

asset_monitoring:attention_required
weather_intelligence:success
demand_forecasting:success
generation_forecasting:success
performance_anomaly:attention_required
battery_management:ready
grid_coordination:success
reviewer:ready_for_human_review